CASCADE: An Onboard Crop Anomaly Screening, Confirmation, and Alert Downlink Engine

For Regenerative Agriculture SmallSat Missions

Emil Wes Lambert, BSc Aerospace Engineering, TU Delft, Solo submission

1 Introduction

Regenerative growers, crop advisors, and food-security operators need to know which field polygons deserve scouting or targeted treatment before stress becomes visible across the whole field. Conventional workflows collect broad scenes, downlink bulk imagery, and wait for ground processing; that can be too slow or bandwidth-heavy for patch-level intervention in systems that aim to avoid blanket chemical application [1, 2, 3, 4]. CASCADE addresses that gap as a hosted SmallSat payload that screens each pass onboard, escalates likely anomalies to confirmation, and exports a compact alert product into farm software. This paper scopes CASCADE to crop anomalies in regenerative agriculture while keeping the onboard architecture reusable for other land-resilience missions.

Table 1: Where CASCADE provides challenge-relevant evidence.

Focus	Where	Key evidence
Onboard autonomy	§2–3	Two-stage screening, confirmation, and priority downlink using NASA-derived indicators.
Feasibility	§2–3	SmallSat resource budget, degradation modes, and real-scene replay.
Impact	§3	Downlink reduction, recall, false-positive, and latency evidence.
Commercial path	§4	Grower alert product, California beachhead, and paid-pilot rollout.
Reproducibility	Eq. (1), Fig. 1	Mission architecture, CSC equation, and reproducible code submission.

2 Proposed Solution Detail

CASCADE's innovation is onboard orchestration, prioritization, and fusion of existing land-observation products under spacecraft resource constraints, not a claim of new crop-stress science. Stage 1 executes an EVI anomaly screen derived from MOD13A1 logic on every observed field patch and updates a per-patch stress posterior using EVI and NDWI cues from the multispectral imager (MSI) pass. When the posterior indicates likely stress, Stage 2 runs a thermal retrieval derived from MOD11A1 and fuses EVI, LST, and MOD09A1-derived NDWI into a Crop Stress Composite (CSC), a normalized anomaly index conceptually related to temperature–vegetation dryness indices (e.g., TVDI) [3, 4, 5, 6]. Those MODIS references define the onboard algorithms; the hosted payload still flies its own MS/TIR sensor stack rather than the MODIS instrument itself. Only confirmed anomalous tiles are buffered for priority downlink.

This makes sensing content-driven rather than timeline-driven. Unlike ASPEN/CASPER-style planners, which optimize scheduled activities, CASCADE schedules per-patch stress belief [7, 8]. It also extends EO-1 ASE's rule-based change-detection heritage with Beta posteriors that accumulate evidence across passes; Dynamic Targeting remains complementary because it focuses on retargeting pointable instruments toward transient scenes rather than composite-cadence crop stress [9, 10].

$$\text{CSC} = 0.578 \cdot \text{clip}\left(\frac{\Delta\text{EVI}}{7.0\sigma_e}, 0, 1\right) + 0.246 \cdot \text{clip}\left(\frac{\Delta\text{LST}}{4.96\sigma_l}, 0, 1\right) + 0.176 \cdot \text{clip}\left(\frac{\Delta\text{NDWI}}{6.0\sigma_n}, 0, 1\right) \qquad (1)$$

where $\Delta\text{EVI} = \max(0, \text{EVI}_{base} - \text{EVI}_{obs})$, $\Delta\text{LST} = \max(0, \text{LST}_{obs} - \text{LST}_{base})$, and $\Delta\text{NDWI} = \max(0, \text{NDWI}_{base} - \text{NDWI}_{obs})$. Each Δ is signed so positive values indicate vegetation loss, thermal rise, or moisture loss relative to baseline. The live CSC weights, saturation divisors, and operating alert threshold ($csc\_alert\_thr = 0.615$) are split-calibrated scheduler parameters from a deterministic two-stage black-box search over fixed 60/20/20 synthetic train/validation/test seed splits that minimizes validation false-positive rate subject to recall and resource guardrails, then promotes only candidates that also clear the held-out test gate; they are not yet crop-specific labeled agronomic weights, which remains a post-TRL-4 retuning task.

The scheduling logic is modeled as a finite-horizon MDP with per-patch Beta posteriors Beta(α_i, β_i) over stress probability that accumulate evidence across passes. Stage 1 increments α_i when EVI or NDWI exceedance is observed and β_i otherwise, with the evidence window capped to preserve responsiveness. The deployed rule-based controller maps patchwise beliefs into a compact summary $(\mu_{\max}, \tau_{\max}, \mathrm{SOC}, d, k)$ for exposition, where $\mu_{\max}$ is the maximum posterior mean, $\tau_{\max}$ is the maximum $P(p > 0.5)$ tail probability, SOC is battery state of charge, $d \in \{0, 1\}$ indicates downlink availability, and k counts passes since last fusion. Action selection uses $\tau_{\max}$, SOC, and k; $\mu_{\max}$ and d enter only the `FUSE`→`FUSE_PRIORITY` promotion gate for alert-tile export under a duty-cycled contact window. The action set remains {SKIP, MOD13, FUSE, FUSE_PRIORITY}; NDWI is derived from the same MSI acquisition rather than requiring a separate action. In the benchmark, the tiered resolution pyramid is modeled as action metadata: MOD13 corresponds to 30 m screening, FUSE corresponds to a 10 m confirmation tier, and FUSE_PRIORITY corresponds to 4.6 m native alert evidence; the 10 m tier is the confirmation-resolution metadata attached to FUSE, not a separate fifth action. The deployed solver is a hand-tuned deterministic policy chosen for flight-V&V traceability.

Table 2: CASCADE engineering targets and demonstrated results.

Target	**Result**
Space latency $\leq$ 90 min	One-pass onboard screening and fusion followed by X-band priority downlink at the first available contact.
End-to-end latency $\leq$ 6 h / $\leq$ 1 h stretch	Read-only overlay delivery after first contact; the stretch case assumes favorable contact timing and downstream integration.
Screening GSD $\leq$ 30 m	Tiered GSD (30/10/4.6 m) tied to MOD13, FUSE, and FUSE_PRIORITY action metadata (§2).
FP rate $\leq 15\%$	0.6% with benign confounders and 36.7 cloud-obscured passes per seed on average ($\sim$31% of 120 passes; §3).
SmallSat resource target	Power: 9.6 W peak / 5.05 W duty-cycled payload average Compute: 92.5% peak LEON4FT ceiling; 39.2% Stage-1; 49.2% seasonal average Energy: 86% observed benchmark minimum SOC in local 40 Wh buffer.
Downlink efficiency	99.1% downlink reduction versus raw (95% CI 99.04–99.07%) and 38.3% energy saving (95% CI 38.20–38.38%) with 100.0% benchmark anomaly-event recall.

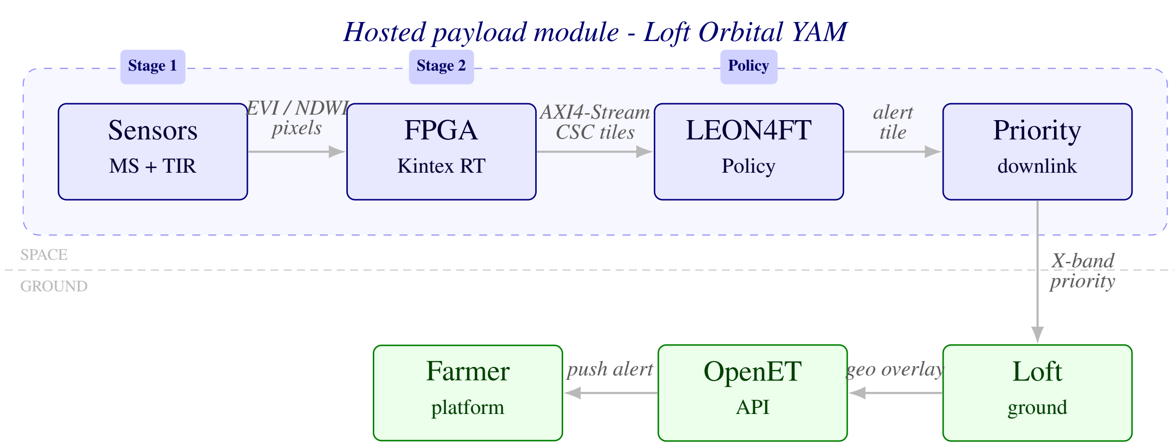

Figure 1: CASCADE end-to-end architecture: hosted payload (dashed) on Loft Orbital YAM from sensors through priority X-band downlink; ground path to farm software via OpenET (full sensing loop, §2).

The spacecraft role is a hosted payload module within a Loft Orbital Yet Another Mission (YAM) bus. The

FPGA performs radiometric preprocessing and per-pixel CSC computation, then transfers frames to the LEON4FT over AXI4-Stream (budgeted under 1 ms). CSC is evaluated as a pass-to-pass relative anomaly detector rather than a fully atmosphere-corrected retrieval. The selected flight part is Kintex UltraScale XQR (XQRKU060) from AMD's radiation-tolerant XQR family [11]; the SWAP budget therefore describes the payload module itself, not the host spacecraft.

Operational degradation is explicit. Between 15% and 35% battery state of charge, CASCADE preserves Stage 1-only screening; below 15%, it drops to alert storage and downlink only. This satisfies the challenge requirement to degrade gracefully under onboard resource stress [1]. Processor utilization remains below the worst-case ceiling and preserves margin for RTOS and fault-handling overhead.

Each alert packet contains a geo-tagged field-patch polygon, CSC severity score, anomaly components, timestamp, pass ID, and a confidence flag from consecutive confirming passes. Escalated alert tiles are compressed onboard using CCSDS 122.0 wavelet coding with a planning assumption of roughly 2.5:1, reducing a nominal 8.4 MB/km^2 native alert stream to about 3.36 MB/km^2 delivered before packetization [12]. Per-patch geolocation is derived from bus ephemeris and attitude solutions and remains inside the 30 m screening-resolution target. Packets move through Loft Orbital's ground network to farm software as read-only geo-tagged overlays via the OpenET-linked API. At TRL 5, CASCADE does not write spray prescriptions back into Climate FieldView or John Deere Operations Center equipment controllers. The latency budget is decomposed into a $\leq$90 min space segment to first ground contact and a $\leq$6 h baseline / $\leq$1 h stretch target from acquisition to appearance in the grower tool.

3 Potential Impact

Cornell seminar material by Dr. Katie Gold emphasizes that spectral stress can emerge before visible symptoms and that early disease-focused anomaly detection improves the odds that intervention is both timely and targeted [2]. Performance is summarized in Table 3. In a 100-seed Monte Carlo benchmark, CASCADE was tested over 120 passes, 500 field patches, 21 benchmark anomaly events, and 9 benign confounders per seed, with 36.7 cloud-obscured passes per seed on average. Across that benchmark, the belief gate routes $\sim$81% of passes to Stage-1 screening only, $\sim$17% to FUSE confirmation, and $\sim$2% to FUSE_PRIORITY alert export.

95% confidence intervals were 99.04–99.07% for downlink reduction versus raw, 38.20–38.38% for energy saving versus raw, 100.0–100.0% for benchmark anomaly-event recall, and 0.43–0.78% for false-positive rate. Recall is the calibration constraint rather than the objective being minimized: the selected operating point minimizes false-positive rate subject to recall and resource guardrails, and the 100.0–100.0% interval confirms that gate remains binding. The meaningful sensitivity story is therefore the threshold sweep: the calibrated operating point is $csc\_alert\_thr = 0.615$, sweeping 0.40 to 0.615 preserved 100.0% recall while moving false-positive rate from 1.52% to 0.61%, at 0.65 recall is 99.3%, and at 0.70 it falls to 84.6%. The matched-recall no-belief ablation reaches 0.5% FP, but at 76.2% seasonal-average compute and 587.9 MB downlink versus 49.2% and 136.3 MB for full CASCADE; the $\pm$30% CSC sweep shows the same calibration dependence, with aggressive high-saturation settings collapsing recall to $\sim$5.4% even as FP trends to zero.

Table 3: Baseline comparisons under the synthetic Monte Carlo benchmark.

Variant	Downlink↓	Recall↑	FP↓	Compute↓
Raw downlink	0.0%	100.0%	0.5%[a]	N/A
Fixed-period onboard	95.8%	100.0%	0.5%	92.5%
EVI-only threshold	96.2%	100.0%	2.3%	39.2%
No-belief CASCADE	95.9%	100.0%	0.5%	76.2%
Full CASCADE	99.1%	100.0%	0.6%	49.2%

[a]Raw downlink FP is the same offline benchmark classifier applied post hoc to the full stream, not an onboard decision.

The real-scene component tests the unchanged scheduler against archival MODIS data spanning the 2014

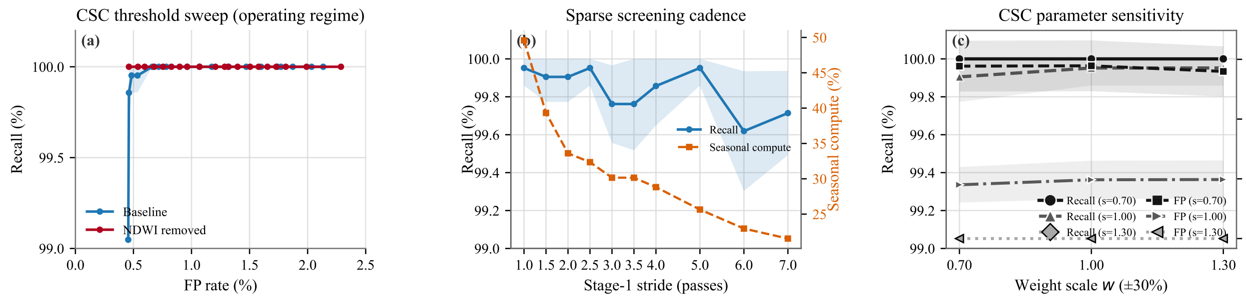

Figure 2: Additional ablations: threshold sweep, sparse Stage 1 cadence, and $\pm 30\%$ CSC sensitivity.

California D4 Exceptional Drought—the most severe event on the USDA Drought Monitor record for the San Joaquin Valley, with the Westlands Water District under 0% surface-water allocation for the full water year [3, 4, 5, 6]. MOD13A1, MOD11A1, and MOD09A1 V061 subsets are pulled via AppEEARS over the Westlands irrigated-farmland core ($36.55°$–$36.65°$ N, $120.45°$–$120.55°$ W) for June through October 2014, aligned to a 1 km grid, collapsed to median clear-sky LST and NDWI over 16-day windows, and replayed with the production-calibrated policy ($csc\_alert\_thr = 0.615$, unmodified). Across 6 valid composite windows after a 3-window warmup, the replay issued **six FUSE_PRIORITY alerts at every active window**: peak CSC 0.869 on 13 August 2014 at 85.2% mean valid coverage, first detection on 28 July—41 days into the active observation window. As a quiet-season complement confirming threshold discipline, the same pipeline over June through October 2024 issued seven FUSE windows, zero FUSE_PRIORITY alerts, and a peak CSC of 0.412 at 99.5% valid coverage, with no parameter adjustment between the two runs.

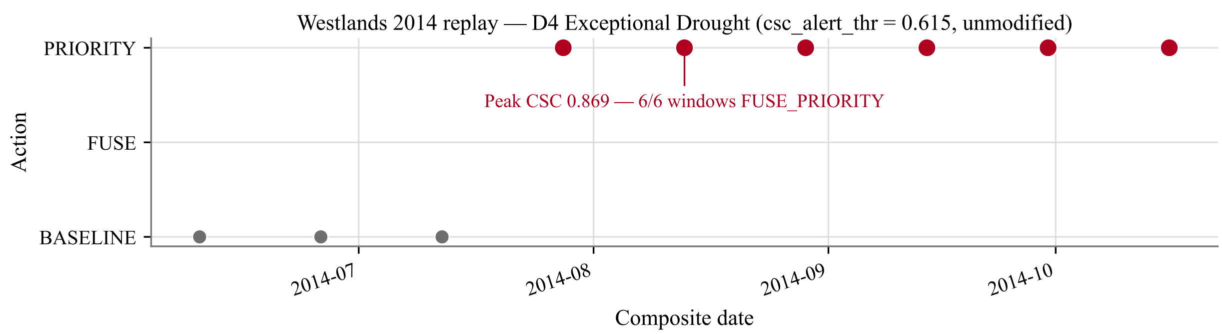

Figure 3: Westlands 2024 quiet-season replay at the production threshold: seven FUSE windows, no priority alert, peak CSC 0.412—the same scheduler that issued six FUSE_PRIORITY alerts across the entire 2014 D4 drought season unchanged.

The farm-scale value proposition is concrete in the disease-focused benchmark scenario. For a 500 ha regenerative tomato operation, UC Cooperative Extension processing-tomato budgets place representative disease-control line items in the rough \$74–119/ha/season range. Under the benchmark assumption that treatment is confined to the roughly 8% of the field flagged during an event, that corresponds to a modeled pilot-farm chemical-spend reduction scenario of roughly \$34,000–\$55,000/season rather than a guaranteed field outcome [13]. That matters because late blight remains a destructive tomato disease that can destroy entire fields and collapse plants within 7–10 days under cool, wet conditions [14]. In broader crop-anomaly monitoring terms, the same onboard triage logic changes how scarce intervention effort is directed, not just farm-level cost [15].

4 Team Experience and Market Evaluation

Emil Wes Lambert is completing a BSc in Aerospace Engineering at TU Delft, with coursework in spacecraft systems, AOCS, mission analysis, and embedded flight software. At AeroDelft he has led partnerships, procurement, and sponsor relations; AeroDelft is a partner in the EUR 73 M HAPSS hydrogen-aviation consortium led by Conscious Aerospace and separately partners with KLM on Project Phoenix. That background supports CASCADE's SWAP realism, AIST-track funding strategy, and commercial positioning. The first commercial system is intentionally lean enough for solo execution because CASCADE sells an onboard scheduler plus alert overlay, not a closed-loop agronomic prescription stack.

The commercialization path below is illustrative rather than predictive. California is the validation anchor and initial go-to-market wedge. The same architecture scales from SJV into the wider U.S. irrigated base, then EU and Brazilian irrigation markets, and finally a global irrigated-farmland platform with region-specific recalibration of CSC thresholds [16, 17, 18, 19]. The economics use a low/design-to-cost case, a $4/ha/year alerting product, 90% annual retention, and a Year 2 soil-carbon MRV add-on aligned to Verra VM0042-style workflows [20]. At the Y5 coverage in Table 4, about 18 Mha, the same benchmark-style treatment-confinement assumption implies an order-of-magnitude of roughly $1–2 B/season in avoidable spend under analogous crop-mix and disease-spend assumptions; this is an impact lens, not a revenue forecast.

Table 4: Illustrative CASCADE rollout from SJV pilot to global platform.

Year	Milestone	Geography	Coverage	Revenue	Op. margin
Y1	SJV pilot	California SJV	3.5% SJV	$0.20 M	grant-funded
Y2	California scale	California	28.3% SJV	$1.70 M	17.1%
Y3	U.S. national	U.S. irrigated	11.0% US	$10.62 M	60.4%
Y4	EU + Brazil entry	EU + Brazil	35.1% EU+Brazil	$27.62 M	68.1%
Y5	Global platform	Global	6.0% global	$76.50 M	70.7%

In this rollout, Y1 is a paid pilot; Y2 scales through Climate FieldView API integration plus direct enterprise sales; Y3 targets U.S.-scale profitability via Climate FieldView and John Deere Operations Center; Y4 expands through xarvio and John Deere Brazil; and Y5 targets global multi-platform distribution that could include CropX. These are API targets, not committed channels. Y1 is modeled as an AIST- or NASA SBIR/STTR-style co-funded bridge-to-MVP year [21, 22]; in the retained fixed-cost sensitivity, the low case breaks even at Y2 and 400 kha, while the base and high cases break even at Y3 and 2.5 Mha.

The near-term product is the grower-facing polygon overlay. The same alert stream could later support parametric-insurance scoring or food-security raster packaging, but those are secondary extensions rather than the initial commercial offer. Agriculture is the entry market, but the scheduler can be retuned for wildfire fuel-moisture monitoring, flood-front tracking, or urban heat-island response by swapping CSC for the relevant anomaly product while reusing the onboard scheduler; the wildfire case alone points toward a market already measured in the tens of billions of dollars [23].

5 Expected Next Steps

CASCADE is currently a TRL 3 concept: the scheduler, split-based CSC calibration, SWAP closure, and real-scene replay are demonstrated in software, but not yet on flight-like compute. The next milestone is a TRL 4 hardware-in-the-loop demonstration on a LEON4FT-class platform under NASA AIST using the replayed EarthData scene set, adding FPGA-side Stage 0 cloud screening. After that, the clearest TRL 5 path is a hosted-payload pilot through Loft Orbital with external validation focused on Y1 pilot assumptions and agronomy review. If selected as a finalist, the immediate pre-pitch actions are a live MODIS replay demo and a first-pass Loft Orbital cost estimate.

6 Conclusion

CASCADE is a reusable onboard triage engine for crop-anomaly monitoring on constrained SmallSat payloads: screening, confirmation, and priority downlink happen onboard, so scarce power, compute, storage, and contact time are spent on the highest-value field patches rather than the full scene. This paper uses regenerative-agriculture crop anomalies as the reference mission case, with a disease-focused benchmark scenario to show agronomic value inside that broader use case. Next is a TRL 4 hardware-in-the-loop demonstration on LEON4FT-class compute, and because the MIT-licensed scheduler is separable from the commercial stack, the same core also supports food-security and humanitarian use.

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